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Week 4 In-Class Activity: Trichomoniasis in a Beef Herd

Before you start

1. Install the packages once (in the RStudio **Console**):

   ```
   install.packages(c("deSolve", "ggplot2", "rmarkdown"))
   ```

2. Run the chunks **in order**, from the top, with the green arrow on each chunk.
3. Each experiment follows the same steps:
   1. read the **explanation** and the **scenario**;
   2. **predict** what will change, *before* calculating anything;
   3. **calculate by hand** with the formulas from the Week 4 lectures;
   4. **run the code** and say whether you were right;
   5. **look at the figure** and explain what it shows.
4. Write your answers after each **Your answer**. In each experiment the line that creates the scenario (`p_new$... <-`) can be changed to test your own ideas.

The result numbers (Proposition 2.3, Theorem 5.1, ...) are the ones on the Week 4 page.

The problem

Bovine trichomoniasis (“trich”) is a sexually transmitted infection of cattle, spread only during breeding. It causes early pregnancy loss, so infected herds have fewer and later calves.

- **Cows** usually clear the infection on their own in about 90 days, and can be infected again: SIS.
- **Bulls**, especially older ones, usually stay infected for the rest of their time in the herd. They leave the infected class only when they are sold or culled.
- Infection passes from an infected bull to a cow very efficiently; bulls are infected by cows much less easily.

This is why Texas requires bulls to be tested, or certified as virgin, before they are sold for breeding (Texas Animal Health Commission).

Our herd. A large ranch with 2,500 cows and 100 bulls, 2,600 animals. The rancher buys **one untested bull**, and it is infected. Time is in **years**.

The groups are the two sexes: bulls (b) and cows (c). Bulls only infect cows and cows only infect bulls, so $\beta_{bb} = \beta_{cc} = 0$. (We use lowercase b and c so they are not confused with the matrix B of the notes.) The transmission numbers are chosen for this activity to be roughly consistent with the facts above; they are not estimates from data.

The model is the two-group SIS model of Section 3 with $\beta_{bb} = \beta_{cc} = 0$:

$$I_b' = (n_b - I_b)\,\beta_{bc}\,I_c - \gamma_b I_b,$$
$$I_c' = (n_c - I_c)\,\beta_{cb}\,I_b - \gamma_c I_c, \qquad n_c = 1 - n_b.$$

Setup: parameters and tools

```
par <- list(
  n_b = 100 / 2600,        # proportion of the herd that is bulls
  beta_bc = 8,             # transmission TO bulls FROM cows (per year)
  beta_cb = 12,            # transmission TO cows FROM bulls (per year)
  gamma_b = 0.25,          # bulls leave the infected class (sold or culled): about 4 years
  gamma_c = 4,             # cows clear the infection: about 1/4 year = 90 days
  seed_b = 1 / 2600,       # at time 0: one infected bull (proportion of the herd)
  seed_c = 0,
  prot_b = 0, prot_c = 0   # fraction of bulls / cows protected (used in Experiment 4)
)
```

These functions are the Week 4 formulas, plus two helpers that print numbers and draw figures. Just run the chunk.

```r
# K = F V^{-1}, with F = diag(n_b, n_c) B and V = diag(gamma_b, gamma_c).
# Groups in the order: bulls, cows.
make_K <- function(p) {
  B <- matrix(c(0,         p$beta_bc,      # row 1: to bulls (from bulls, from cows)
                p$beta_cb, 0),             # row 2: to cows  (from bulls, from cows)
              nrow = 2, byrow = TRUE)
  S0 <- c((1 - p$prot_b) * p$n_b, (1 - p$prot_c) * (1 - p$n_b))  # susceptibles at P0
  F <- diag(S0) %*% B
  V <- diag(c(p$gamma_b, p$gamma_c))
  F %*% solve(V)
}

# Jacobian at the disease-free equilibrium: J(P0) = F - V
make_J0 <- function(p) {
  V <- diag(c(p$gamma_b, p$gamma_c))
  make_K(p) %*% V - V
}

# spectral radius: the largest absolute value of the eigenvalues
spec_rad <- function(A) max(Mod(eigen(A)$values))

# simulate the model; output in proportions of the herd
simulate <- function(p, years = 30) {
  model <- function(t, y, p) {
    Ib <- y[1]; Ic <- y[2]
    Sb <- (1 - p$prot_b) * p$n_b - Ib          # susceptible bulls (protected ones excluded)
    Sc <- (1 - p$prot_c) * (1 - p$n_b) - Ic    # susceptible cows
    dIb <- Sb * p$beta_bc * Ic - p$gamma_b * Ib
    dIc <- Sc * p$beta_cb * Ib - p$gamma_c * Ic
    list(c(dIb, dIc))
  }
  out <- as.data.frame(ode(y = c(p$seed_b, p$seed_c), times = seq(0, years, by = 0.02),
                           func = model, parms = p, rtol = 1e-10, atol = 1e-15))
  names(out) <- c("time", "Ib", "Ic")
  out
}

# the numbers we compare in every experiment
numbers <- function(p) {
  end <- tail(simulate(p), 1)
  c(R0 = spec_rad(make_K(p)),
    growth_rate_r = max(eigen(make_J0(p))$values),
    infected_bulls_after_30y = round(2600 * end$Ib, 1),
    infected_cows_after_30y  = round(2600 * end$Ic, 1))
}

# figure: infected animals over time, for several scenarios
compare <- function(scenarios, years = 30, log_scale = FALSE) {
  d <- do.call(rbind, lapply(names(scenarios), function(nm) {
    out <- simulate(scenarios[[nm]], years)
    data.frame(time = out$time, infected = 2600 * (out$Ib + out$Ic), scenario = nm)
  }))
  d$scenario <- factor(d$scenario, levels = names(scenarios))
```

```
  g <- ggplot(subset(d, infected > 0), aes(time, infected, colour = scenario, linetype = scenar
io)) +
    geom_line(linewidth = 1.1) +
    scale_colour_manual(values = c("#000000", "#D55E00", "#0072B2", "#009E73")) +
    labs(x = "years", y = "infected animals (herd of 2,600)", colour = NULL, linetype = NULL)
  if (log_scale) g <- g + scale_y_log10()
  g
}
```

Warm-up: the original herd

Theory (Section 4, Step 6; Proposition 2.3; Theorem 5.1). With $\beta_{bb} = \beta_{cc} = 0$,

$$K = \begin{pmatrix} 0 & k_{bc} \\ k_{cb} & 0 \end{pmatrix}, \qquad k_{bc} = \frac{\beta_{bc} n_b}{\gamma_c}, \quad k_{cb} = \frac{\beta_{cb} n_c}{\gamma_b},$$

where k_{ij} is the number of new infections in group i caused by one infected animal of group j.

Putting $k_{bb} = k_{cc} = 0$ in equation (8) gives

$$R_0 = \sqrt{k_{bc}\, k_{cb}}.$$

The Jacobian at the disease-free equilibrium is $J(P_0) = F - V = \begin{pmatrix} -\gamma_b & \beta_{bc} n_b \\ \beta_{cb} n_c & -\gamma_c \end{pmatrix}$, and the growth rate r is its largest eigenvalue (Section 5).

Calculate by hand:

1. k_{bc} and k_{cb}. Why is one so much larger than the other?
2. R_0. Will trich take hold in the herd?
3. The trace and determinant of $J(P_0)$, and r from $r = \dfrac{\mathrm{tr}J + \sqrt{(\mathrm{tr}J)^2 - 4\det J}}{2}$. (Since $\gamma_b \neq \gamma_c$, the shortcut $r = \gamma(R_0 - 1)$ does **not** apply here.)

Your answer: 1. $k_{bc} = \frac{\beta_{bc} n_b}{\gamma_c} = \frac{8 * \frac{1}{26}}{4} = \frac{1}{13}$

$k_{cb} = \frac{\beta_{cb} n_c}{\gamma_b} = \frac{12 * \frac{25}{26}}{\frac{1}{4}} = \frac{600}{13}$

There are more susceptable cows than bulls at any given time, bulls stay infective for longer than cows do, giving them more chances to infect in general. Cows also get infected at a higher rate than bulls.

2. $R_0 = \sqrt{k_{bc} k_{cb}} = \sqrt{\frac{600}{13^2}} \approx 1.88$

The trich will take hold in the herd.

3. $Tr(J) = -\gamma_b - \gamma_c = -\frac{1}{4} - 4 = -\frac{17}{4}$

$$Det(J) = (-\gamma_b)(-\gamma_c) - (\beta_{bc} n_b)(\beta_{cb} n_c) = (\frac{-1}{4})(-4) - (8 * \frac{1}{26})(12 * \frac{25}{26}) = 1 - \frac{600}{169} = \frac{-431}{169}$$

$$r = \frac{\mathrm{tr}J + \sqrt{(\mathrm{tr}J)^2 - 4\mathrm{det}J}}{2} = \frac{\frac{-17}{4} + \sqrt{(\frac{-17}{4})^2 - 4(\frac{-431}{169})}}{2} \approx 0.533$$

```
make_K(par)
```

```
##          [,1]       [,2]
## [1,]  0.00000 0.07692308
## [2,] 46.15385 0.00000000
```

```
make_J0(par)
```

```
##          [,1]       [,2]
## [1,] -0.25000  0.3076923
## [2,] 11.53846 -4.0000000
```

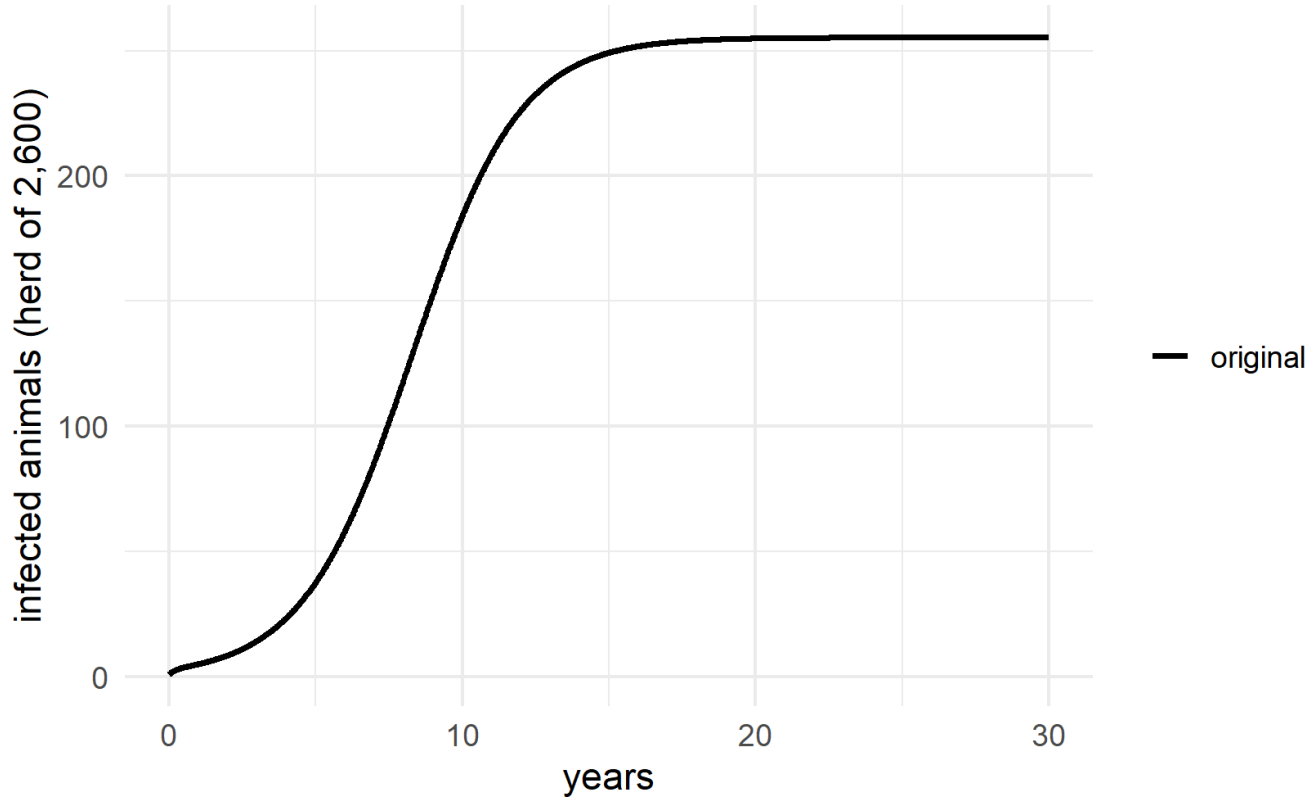
```
numbers(par)
```

```
##                       R0            growth_rate_r infected_bulls_after_30y
##                 1.884223                 0.533180                 69.600000
##  infected_cows_after_30y
##                185.800000
```

Were you right?

Your answer: Yes, I was right.

```
compare(list(original = par))
```



What does the figure show? How many years until the number of infected animals levels off?

Your answer: The figure shows the number of infected animals over time in the herd, mesured in years. It levels off in roughly 15-20 years.

Experiment 1. Test and cull bulls every year

Explanation. Testing bulls every year and culling the infected ones means a bull stays infected in the herd for about 1 year instead of 4: γ_b goes from 0.25 to 1.

Predict (before calculating): will R_0 go up or down? Will trich still take hold?

Your answer: Prediction: R_0 will decrease to the point that the trich will not take hold.

Calculate by hand: the new k_{cb} and the new R_0. Then find the **critical** γ_b, where $R_0 = 1$ (set $k_{bc}k_{cb} = 1$ and solve for γ_b). How long can an infected bull stay in the herd before trich takes hold?

Your answer: $k_{cb} = \frac{\beta_{cb} n_c}{\gamma_b} = \frac{12\frac{25}{26}}{1} = \frac{150}{13}$

$R_0 = \sqrt{\frac{150}{13^2}} \approx 0.942$

$R_0 = 1 = \sqrt{k_{bc}k_{cb}}\ 1 = k_{bc}k_{cb} --> 13 = k_{cb} = \frac{\beta_{cb} n_c}{\gamma_b}\gamma_b = \frac{\beta_{cb} n_c}{13} = \frac{12\frac{25}{26}}{13} \approx 0.8876$

$\frac{1}{\gamma_b} = 1.12666...years$

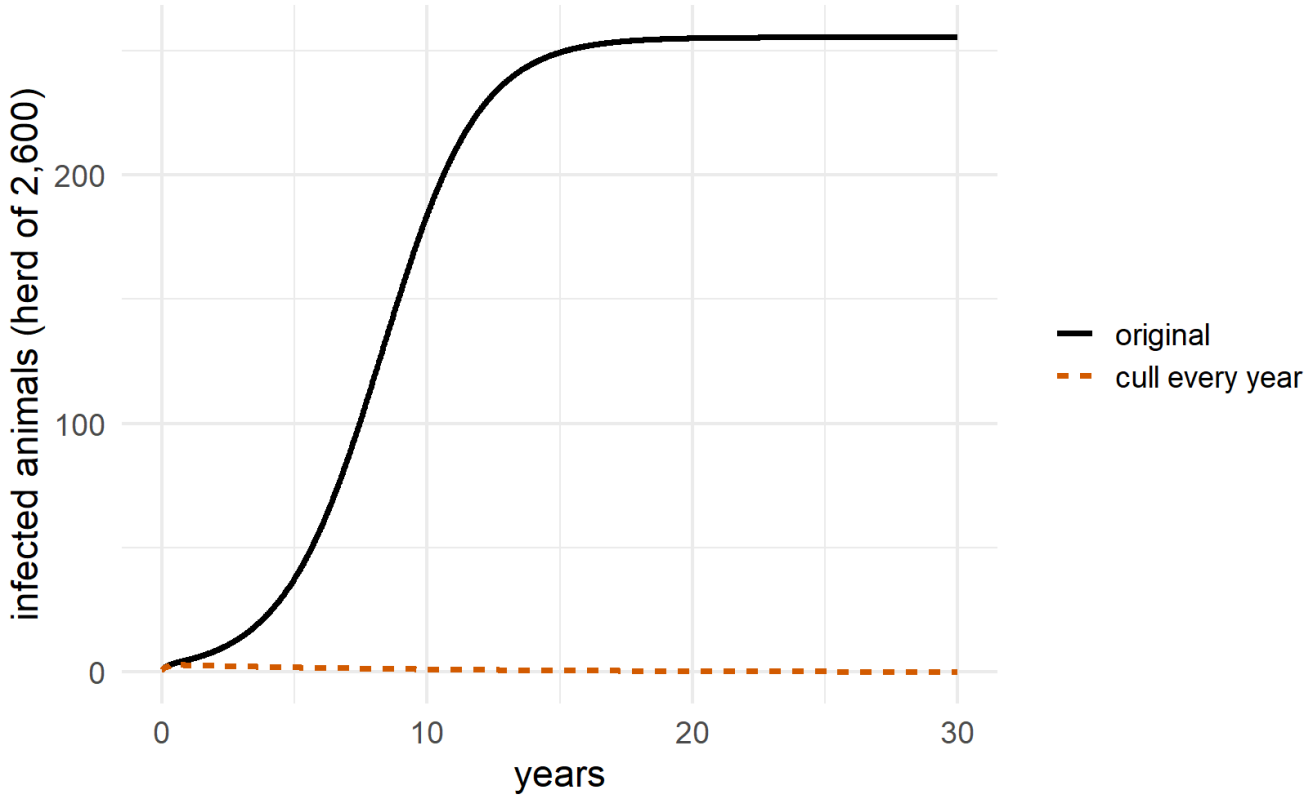
```
p_new <- par
p_new$gamma_b <- 1                      # the scenario: change this to test other values
rbind(original = numbers(par), cull_every_year = numbers(p_new))
```

```
##                       R0 growth_rate_r infected_bulls_after_30y
## original        1.8842229    0.53317999                    69.6
## cull_every_year 0.9421114   -0.09161966                     0.0
##                 infected_cows_after_30y
## original                          185.8
## cull_every_year                     0.1
```

Were you right?

Your answer: I was.

```
compare(list(original = par, "cull every year" = p_new))
```



What does the figure show? Why?

Your answer: The figure shows that if the bulls are culled yearly, the disease is unable to take hold of the herd.

Experiment 2. Same R_0, different speed?

Explanation. Two ranchers each double one rate. Rancher A removes infected bulls twice as fast (γ_b from 0.25 to 0.5). Rancher B treats cows so they clear trich twice as fast (γ_c from 4 to 8).

Predict: which rancher lowers R_0 more? Which one slows the spread more?

Your answer: Prediction: They should be the same?

Calculate by hand: R_0 for each rancher. (Hint: how do γ_b and γ_c enter $R_0 = \sqrt{k_{bc}k_{cb}}$?) Then r for each one, from the trace and determinant of their $J(P_0)$.

Your answer: Rancher A: $k_{bc} = \frac{8 * \frac{1}{26}}{4} = \frac{1}{13}\, k_{cb} = \frac{12 * \frac{25}{26}}{\frac{1}{2}} = \frac{300}{13}$

Rancher B: $k_{bc} = \frac{8 * \frac{1}{26}}{8} = \frac{1}{26}\, k_{cb} = \frac{12 * \frac{25}{26}}{\frac{1}{4}} = \frac{600}{13}$

$R_{0A} = \sqrt{\frac{300}{13} * \frac{1}{13}} \approx 1.33\; R_{0B} = \sqrt{\frac{600}{13} * \frac{1}{26}} \approx 1.33$

$Tr(J_A) = \frac{-1}{2} - 4 = -4.5\; Det(J_A) = (\frac{-1}{2})(-4) - (8 * \frac{1}{26})(12 * \frac{25}{26}) = 2 - \frac{600}{169} = \frac{-262}{169}$ $r_A = $

$Tr(J_B) = \frac{-1}{4} - 8 = -8.25\; Det(J_B) = (\frac{-1}{4})(-8) - (8 * \frac{1}{26})(12 * \frac{25}{26}) = 2 - \frac{600}{169} = \frac{-262}{169}$ $r_B = $

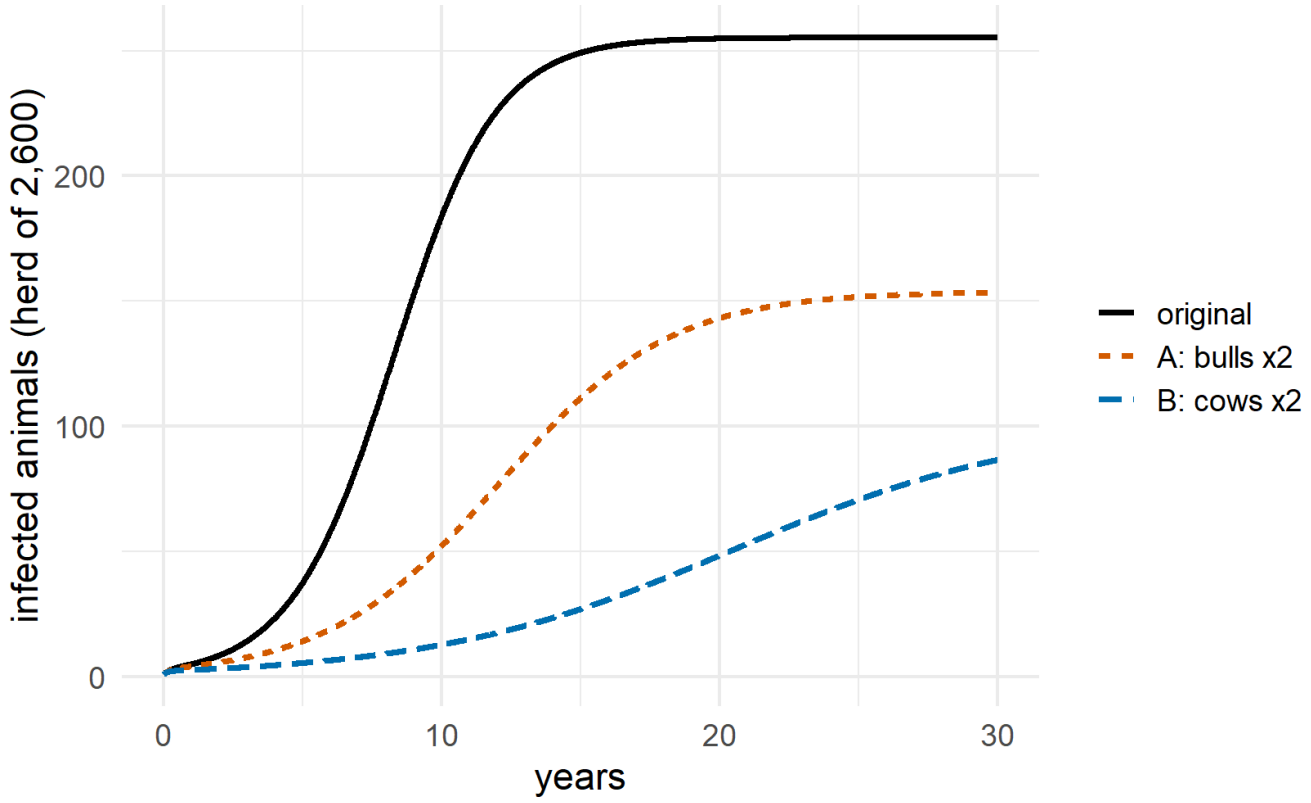
```
p_A <- par; p_A$gamma_b <- 0.5          # rancher A: bulls
p_B <- par; p_B$gamma_c <- 8            # rancher B: cows
rbind(original = numbers(par), rancher_A = numbers(p_A), rancher_B = numbers(p_B))
```

```
##                R0 growth_rate_r infected_bulls_after_30y
## original  1.884223     0.5331800                     69.6
## rancher_A 1.332347     0.3215357                     40.9
## rancher_B 1.332347     0.1838190                     36.0
##           infected_cows_after_30y
## original                    185.8
## rancher_A                   112.6
## rancher_B                    50.7
```

Were you right? Why can two herds have the **same** R_0 but different growth rates r?

Your answer: I wasn't right, because while they have the same R_0, reducing the infectious time for the larger population seems to be more effective in reducing infection in the herd.

```
compare(list(original = par, "A: bulls x2" = p_A, "B: cows x2" = p_B))
```



What does the figure show? Compare how fast each herd reaches its final level, and the final levels themselves.

Your answer: The figure shows that while both strategies make the herd take longer to get infected, and to resolve to a lower level, treating the cows causes significantly lower infection rates than culling bulls twice as often.

Experiment 3. What if trich arrives in a cow?

Explanation. Instead of an infected bull, the rancher buys one infected cow.

Predict: does R_0 change? Does the growth rate r change? Does the start of the epidemic look different?

Your answer:

Calculate by hand: which of $k_{bc}, k_{cb}, R_0, \mathrm{tr}J(P_0)$ and $\det J(P_0)$ depend on where the infection starts? Then, at $t = 0$, with $I_b = 0$ and I_c equal to one cow, use the model equations: is I_c going up or down? And I_b?

Your answer:

```
p_new <- par
p_new$seed_b <- 0
p_new$seed_c <- 1 / 2600                # the scenario: one infected cow
rbind(original = numbers(par), starts_in_a_cow = numbers(p_new))
```

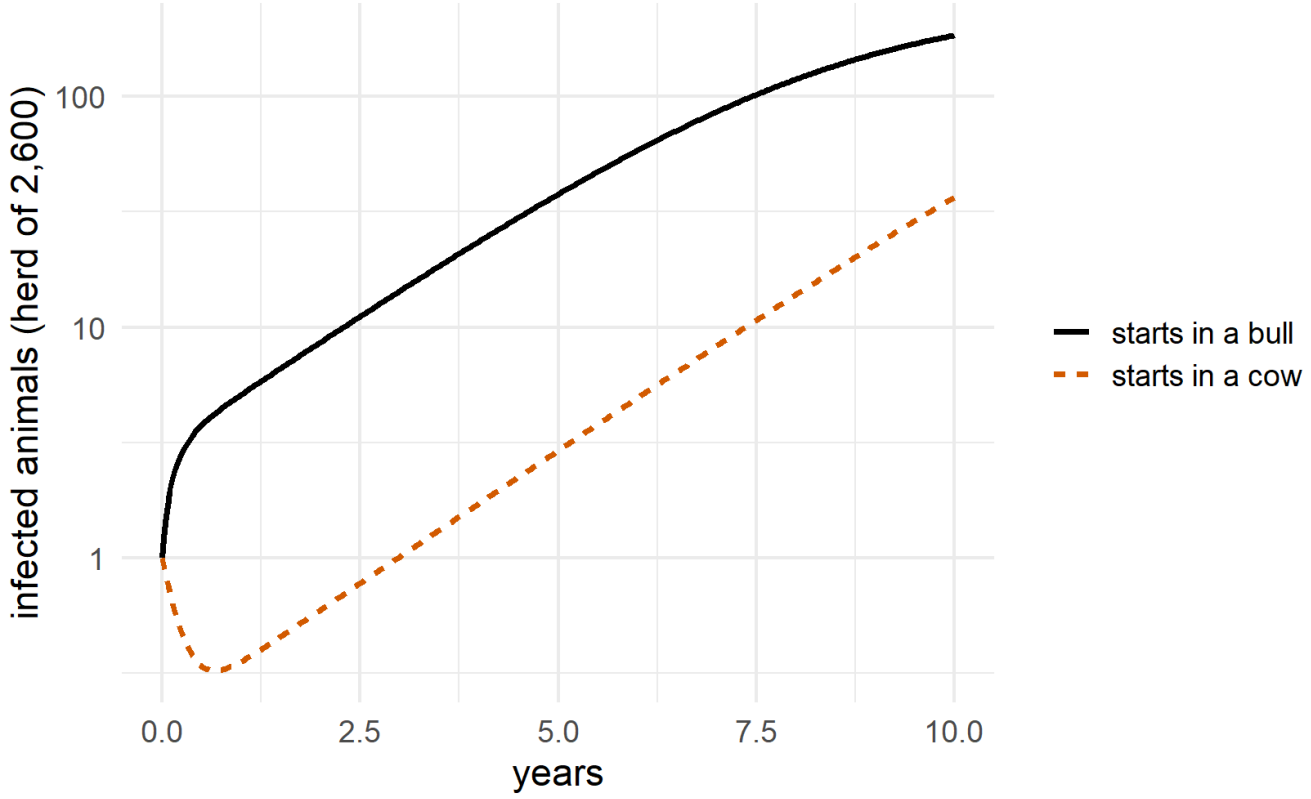
```
##                        R0 growth_rate_r infected_bulls_after_30y
## original        1.884223       0.53318                     69.6
## starts_in_a_cow 1.884223       0.53318                     69.6
##                 infected_cows_after_30y
## original                          185.8
## starts_in_a_cow                   185.8
```

Were you right?

Your answer: $k_{bc}, k_{cb}, R_0, \mathrm{tr}J(P_0)$ and $\detJ(P_0)$ do not depend on where the infection starts in the group.

$$I_b' = (n_b - I_b)\,\beta_{bc}\,I_c - \gamma_b I_b,$$
$$I_c' = (n_c - I_c)\,\beta_{cb}\,I_b - \gamma_c I_c, \qquad n_c = 1 - n_b.$$

```
compare(list("starts in a bull" = par, "starts in a cow" = p_new), years = 10, log_scale = TRU
E)
```



What does the figure show? Why does the cow start drop at first, and why do the two curves end up parallel? (Section 5: after the transient, every start grows like e^{rt}.)

Your answer: The figure shows an initial drop, because transmission from a cow is harder than from a bull, however once a bull is infected, it is functionally no different from the original scenario with a time delay, hence why it becomes parallel with the original. ————————————————————————

Experiment 4. Who should be protected?

Explanation. A thought experiment: imagine a vaccine that protects an animal completely and can be given to bulls or to cows. (Real trich vaccines are given to cows and only reduce infection.) The rancher has **75 doses**. Protected animals can never be infected, so protecting fractions p_b of the bulls and p_c of the cows lowers the susceptibles at P_0 to $(1-p_b)n_b$ and $(1-p_c)n_c$, which multiplies the rows of K by $1-p_b$ and $1-p_c$ (Section 7):

$$K_p = \begin{pmatrix} 0 & (1-p_b)\,k_{bc} \\ (1-p_c)\,k_{cb} & 0 \end{pmatrix}.$$

Predict: where should the 75 doses go: bulls, cows, or spread at random?

Your answer: Prediction: Because treating the cows was more effective in experiment two, vaccinating the cows should help.

Calculate by hand: show that $R_p = \sqrt{(1-p_b)(1-p_c)}\;R_0$. Then compute R_p for each plan:

- **bulls only:** $p_b = 75/100, p_c = 0$;
- **cows only:** $p_b = 0, p_c = 75/2500$;
- **at random:** $p_b = p_c = 75/2600$.

Your answer: $k_{bc} = \frac{\beta_{bc} n_b (1-p_b)}{\gamma_c}$

$k_{cb} = \frac{\beta_{cb} n_c (1-p_c)}{\gamma_b}$

$R_p = \sqrt{k_{bc}k_{cb}} = \sqrt{\frac{\beta_{cb} n_b (1-p_b)\beta_{bc} n_c (1-p_c)}{\gamma_c \gamma_b}} = \sqrt{(1-p_b)(1-p_c)}\sqrt{\frac{\beta_{cb} n_b \beta_{bc} n_c}{\gamma_c \gamma_b}} = \sqrt{(1-p_b)(1-p_c)}R_0$

Bulls only: $p_b = \frac{75}{100}\, p_c = 0\; R_p = \sqrt{(\frac{25}{100})(1)}R_0 = \frac{R_0}{2} \approx 0.94$

Cows only: $p_b = 0\, p_c = \frac{75}{2500}\, R_p = \sqrt{(\frac{2425}{2500})(1)}R_0 \approx R_0 * 0.98489 \approx 1.856$

Random: $p_b = \frac{75}{2600}\, p_c = \frac{75}{2600}\, R_p = \sqrt{(\frac{2425}{2600})(\frac{2425}{2600})}R_0 \approx 1.8299$

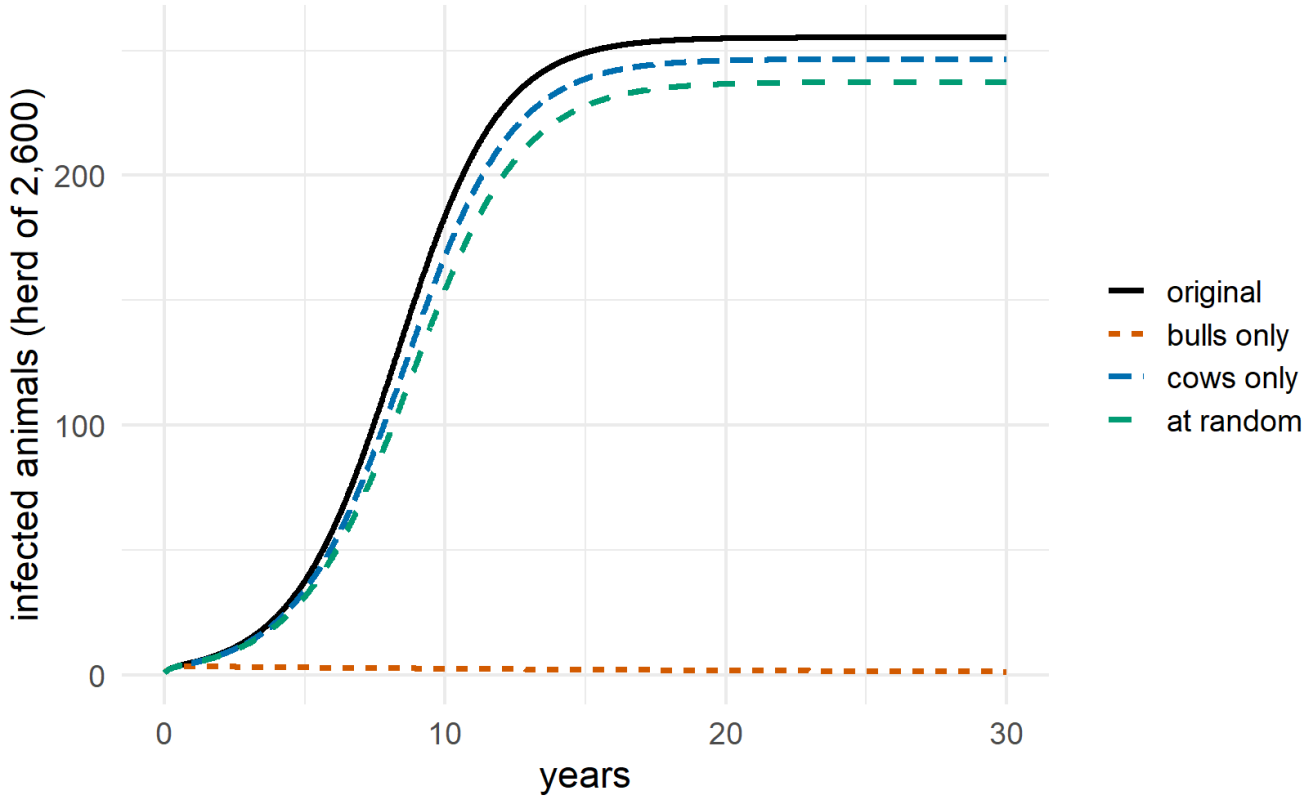
```
protect <- function(p, pb, pc) { q <- p; q$prot_b <- pb; q$prot_c <- pc; q }
p_bulls  <- protect(par, pb = 75 / 100, pc = 0)
p_cows   <- protect(par, pb = 0,        pc = 75 / 2500)
p_random <- protect(par, pb = 75 / 2600, pc = 75 / 2600)
rbind(original = numbers(par), bulls_only = numbers(p_bulls),
      cows_only = numbers(p_cows), at_random = numbers(p_random))
```

```
##                  R0 growth_rate_r infected_bulls_after_30y
## original   1.8842229    0.53317999                     69.6
## bulls_only 0.9421114   -0.02661992                      0.4
## cows_only  1.8557443    0.51306975                     68.7
## at_random  1.8298703    0.49493326                     65.9
##            infected_cows_after_30y
## original                     185.8
## bulls_only                     1.1
## cows_only                    178.0
## at_random                    171.6
```

Were you right? Why does the same number of doses have such different effects?

Your answer: The disease can only be transmitted by cows to bulls and from bulls to cows, so vaccinating 3/4ths the bull population stalls out the number of susceptable transmissions.

```
compare(list(original = par, "bulls only" = p_bulls,
             "cows only" = p_cows, "at random" = p_random))
```



What does the figure show? How many doses would be needed if they all went to cows? (Hint: set $R_p = 1$ with $p_b = 0$.)

Your answer: The figure shows the similarity of levels with only cows vaccinated and random vaccinations, and how the disease doesn't take hold in the herd with only bulls vaccinated.

$$R_p = 1\sqrt{(1)(1-p_c)}\,1 - \frac{1}{R_0^2} = p_c = 1 - \frac{13^2}{600} = \frac{431}{600} - - > \ 1796\ Doses$$

Experiment 5. Artificial insemination

Explanation. Breeding a fraction a of the cows by artificial insemination (AI) removes those matings, so both β_{bc} and β_{cb} are multiplied by $1 - a$.

Predict: if 30% of the cows are bred by AI, does trich disappear? And with 50%?

Your answer: Prediction: 30% will not be enought, but 50% will be.

Calculate by hand: show that R_0 is multiplied by $1 - a$. Compute R_0 for $a = 0.3$ and $a = 0.5$, and the smallest a that stops trich.

Your answer: $R_{0a} = \sqrt{\frac{\beta_{bc}(1-a)n_b\beta_{cb}(1-a)n_c}{\gamma_c\gamma_b}} = \sqrt{(1-a)^2} \sqrt{\frac{\beta_{bc}n_b\beta_{cb}n_c}{\gamma_b \gamma_c}} = (1-a) R_0}$

a=0.3 $R_0a = (1-0.3)R_0 $

a=0.5 $R_0a = (1-0.5)R_0 $

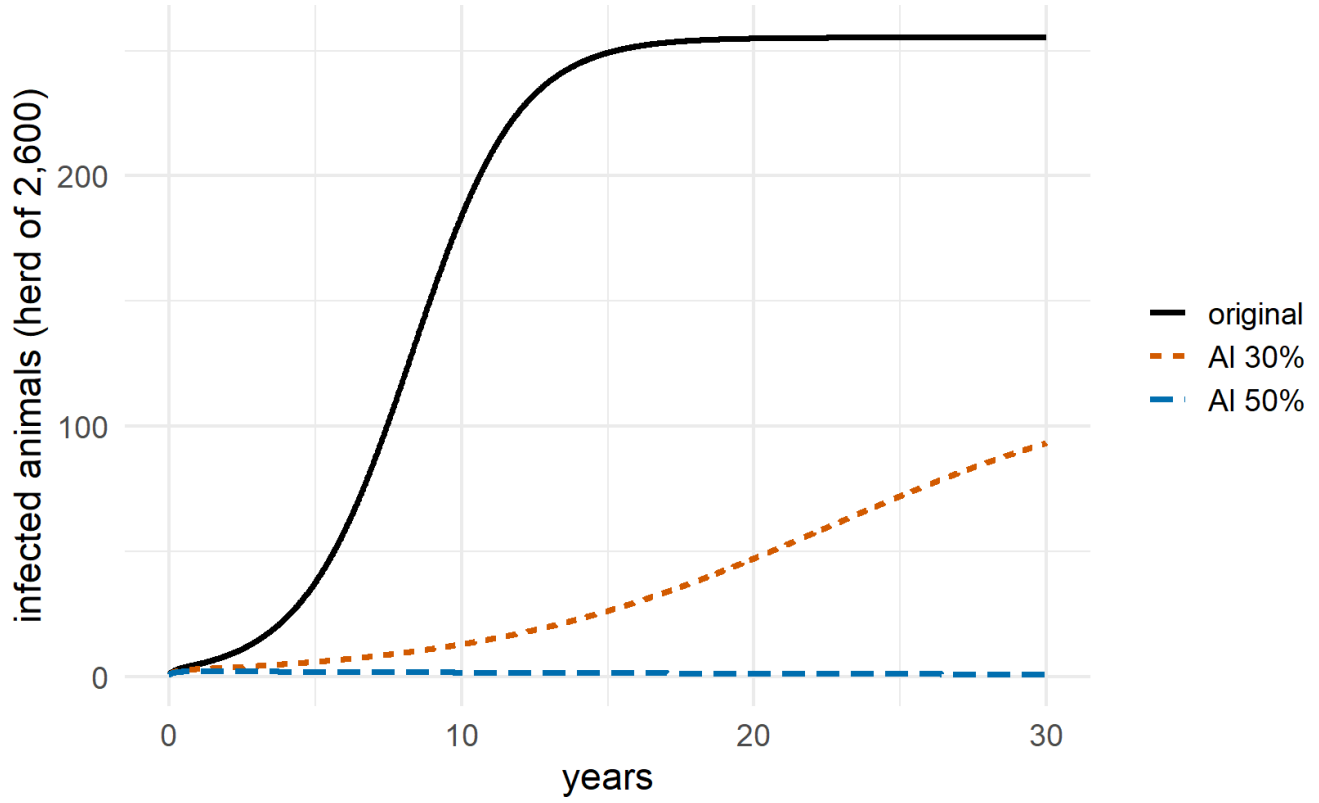
```
ai <- function(p, a) { q <- p; q$beta_bc <- (1 - a) * p$beta_bc; q$beta_cb <- (1 - a) * p$beta_
cb; q }
p_30 <- ai(par, 0.3)
p_50 <- ai(par, 0.5)
rbind(original = numbers(par), AI_30 = numbers(p_30), AI_50 = numbers(p_50))
```

```
##                 R0 growth_rate_r infected_bulls_after_30y
## original 1.8842229    0.53317999                     69.6
## AI_30    1.3189560    0.16743756                     31.6
## AI_50    0.9421114   -0.02661992                      0.4
##          infected_cows_after_30y
## original                   185.8
## AI_30                       61.7
## AI_50                        0.6
```

Were you right?

Your answer: I was correct!

```
compare(list(original = par, "AI 30%" = p_30, "AI 50%" = p_50))
```



What does the figure show? With 30% AI, trich still persists. Is that outcome still useful to the rancher? Why?

Your answer: Even if the disease persists, the stable level of infection in the herd is significantly lower than the original.

Wrap-up

In two or three sentences each:

1. Neither bulls alone nor cows alone could keep trich going. How can the two together have $R_0 > 1$?
2. Which experiment surprised you most? What had you predicted, and what happened instead?
3. If you were advising this rancher, which measure would you recommend first, and why?

Your answer: 1. the disease can only transfer between the two groups.

2. Experiment 2, with the same R_0 not meaning the same growth r.
3. Vaccinating the bulls in the herd, because it seems to be the least disruptive to the ranch's schedule, and completely eliminates the trich.

Sources for the facts about trich: Mississippi State University Extension, *Trichomoniasis in Beef Cattle*; MSD Veterinary Manual, *Trichomoniasis in Cattle*; Texas Animal Health Commission, *Texas Trichomoniasis Program*.